Rocket Science and Technology 4363 Motor Ave., Culver City, CA 90232
Phone: (310) 839-8956 Fax: (310) 839-8855

Upper Stage Bending Influence on Static Stability(rev.3)

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by C.P.Hoult

- The Nike-Cajun Problem

Introduction

This problem began in 1956. At that time, the Nike-boosted Cajun, or Nike Cajun, was just entering Air Force service as a small, inexpensive sounding rocket. The Nike Cajun concept originated with the Univ. of Michigan (U. of M.) which, in those days, was far from a systems house as we would understand the term today. They approached rocketry from an empirical perspective: let's launch one and see what happens. They did not bother to do trajectory analysis, or heating and stress calculations.

The first results were spectacularly pleasant: The Nike Cajun was inexpensive, and it often seemed to work. Only later did its warts become apparent. For example, in the beginning, there was no easy way to adjust the second stage fins to obtain a desired roll rate. To overcome this deficiency, roll wedges were attached to the fin trailing edges. But the wedges caused so much fin twisting moment that roll rate prediction was often accompanied by prayer. We later discovered that the baseline U. of M. fin design, when wedged and flown with a very light payload (high burnout Mach No. and dynamic pressure), came perilously close to aileron reversal.

All these little problems, however, paled beside the real excitement. When launched with a long, heavy payload, the rocket would often abruptly fail structurally[1] immediately prior to first stage burnout. Camera data showed the entire second stage progressively bending with the payload eventually snapping off. The immediate, ad hoc, solution was to beef up the payload-to-second stage joint. This helped a little, but failed to solve the problem. Eventually, calculations (I recall Bob Swanson (Aerojet) as the first to propose an aeroelastic explanation) similar to those replicated here established static aeroelasticity as the causative mechanism.

Analysis

Consider a non-rolling two stage rocket as sketched in Figure 1. Suppose that there is a thrust misalignment that produces a quasi steady state angle of attack.

Define the notation:

$\mathbf{N_{T1}}$ = First stage tail normal force,
$\mathbf{N_{T2}}$ = Second stage tail + interstage normal force,
$\mathbf{N_N}$ = Nose normal force,
U = Free stream velocity,

[1] Sometimes called a RUD, or Rapid Unscheduled Disassembly, event

α = Angle of attack,
T = Booster thrust,
δ = Thrust misalignment angle, and
x's as sketched in Fig. 1.

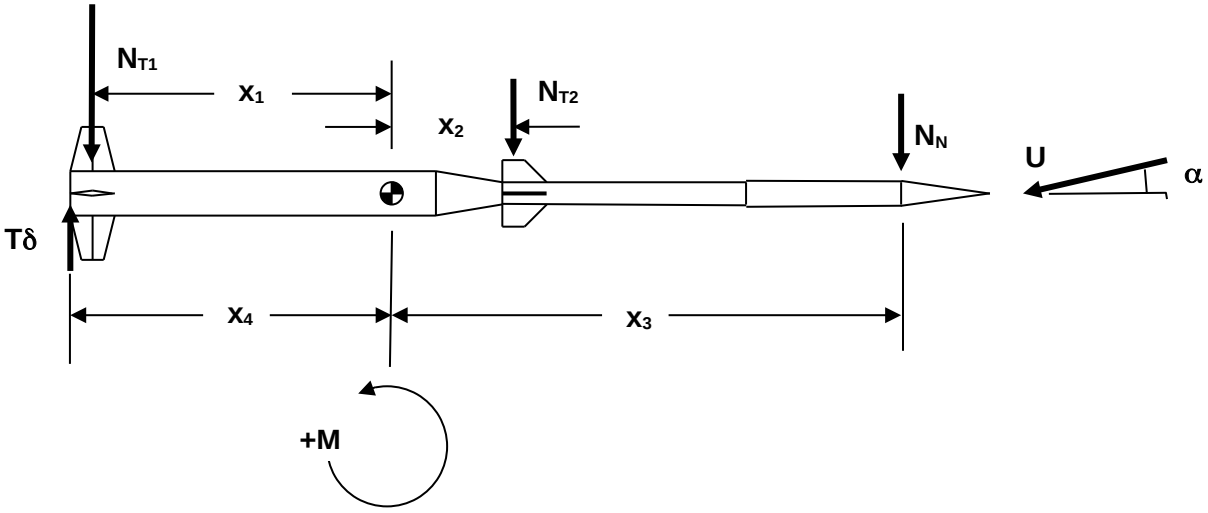


Figure 1

In the steady state, the net moment must vanish. Using the moment sign convention sketched in Fig. 1,

$$N_{T1}\, x_1 - N_{T2}\, x_2 - N_N\, x_3 - T\delta\, x_4 = 0.$$

It is straightforward to relate the aerodynamic normal force components to the angle of attack. For a rigid airframe,

$$N_{T1} = q\, S\, C_{N1\alpha}\, \alpha$$

$$N_{T2} = q\, S\, C_{N2\alpha}\, \alpha, \text{ and}$$

$$N_N = q\, S\, C_{NN\alpha}\, \alpha, \text{ where}$$

q = Dynamic pressure,
S = Aerodynamic reference area, and
$C_{N\alpha}$ = Normal force coefficient slope

Now, suppose the first stage booster is so stiff that its deformations under load are negligible. Under this assumption, we can model the flexible second as though it were cantilevered from the booster. The third of the aerodynamic load equations above must be modified to reflect its bending deflection, **y**. See Figure 2 below:

$$N_N = q\, S\, C_{NN\alpha}\, (\alpha + dy / dx).$$

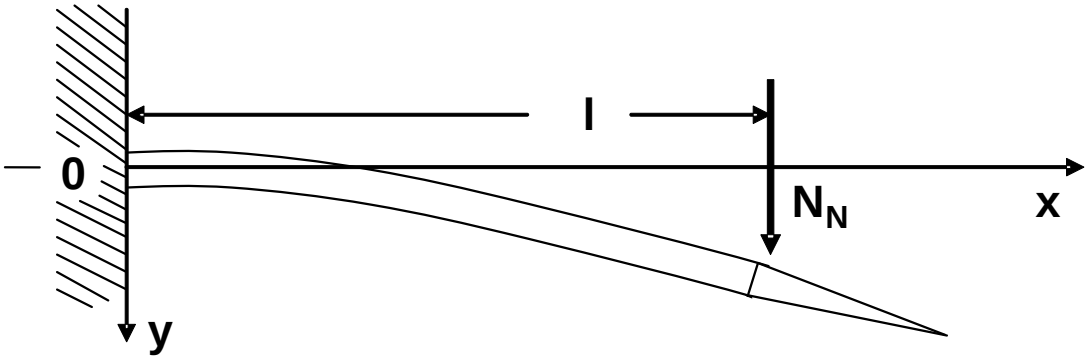


Figure 2

Next, the slender beam equation is, neglecting lateral and axial acceleration loads,

$$M = N_N (l - x) = EI_2 \, d^2y / dx^2, \text{ where}$$

M = Bending moment,

l = Distance from nose tip center of pressure to the base of the upper stage/dart,

EI_2 = Second stage bending stiffness $\approx E \pi R^3 t$,

where R = Radius of a thin-walled tube, and

t = Thickness of a thin walled-tube.

After integrating once,

$$EI_2 \, dy / dx = N_N (l x - \tfrac{1}{2} x^2), \text{ or}$$

$$dy / dx (l) = q S C_{NN\alpha} (\alpha + dy / dx (l)) l^2 / 2 EI_2, \text{ or}$$

$$dy / dx (l) = q S C_{NN\alpha} \, \alpha \, l^2 / (2 EI_2 - q S C_{NN\alpha} \, l^2).$$

When the denominator above vanishes, it leads to an infinite system response called divergence. It is rightly feared because it defines an important system limitation.

Now, return to the condition of system moment balance. With aeroelastic effects included, we have that

$$q S \alpha \{ x_1 C_{N1\alpha} - x_2 C_{N2\alpha} - x_3 C_{NN\alpha} [2 EI_2 / (2 EI_2 - q S C_{NN\alpha} l^2)] \} - x_4 T\delta = 0.$$

The angle of attack is immediately found to be

$$\alpha \;=\; x_4\, T\delta \;/\; q\, S \left\{ x_1\, C_{N1\alpha} - x_2\, C_{N2\alpha} - x_3\, C_{NN\alpha} \left[2\, EI_2 \,/\, (2\, EI_2 - q\, S\, C_{NN\alpha}\, l^2) \right] \right\}.$$

Next, define the divergence dynamic pressure, q_D, to be

$$q_D \;=\; 2\, EI_2 \,/\, S\, C_{NN\alpha}\, l^2.$$

Then, using this result, the angle of attack becomes

$$\alpha \;=\; x_4\, T\delta \;/\; q\, S \left\{ x_1\, C_{N1\alpha} - x_2\, C_{N2\alpha} - x_3\, C_{NN\alpha}\, q_D \,/\, (q_D - q) \right\}.$$

The first effect of static aeroelasticity is that the angle of attack on all elements of the body is increased:

$$\alpha \;=\; \alpha_{Rigid} \left\{ x_1\, C_{N1\alpha} - x_2\, C_{N2\alpha} - x_3\, C_{NN\alpha} \right\} / \left\{ x_1\, C_{N1\alpha} - x_2\, C_{N2\alpha} - x_3\, C_{NN\alpha}\, q_D \,/\, (q_D - q) \right\} \quad (1)$$

The nose tip slope is

$$dy / dx\,(l) = q\, \alpha \;/\; (q_D - q).$$

The total nose aerodynamic force is

$$N_N \;=\; q\, S\, C_{NN\alpha} \left(\alpha + dy / dx \right) \;=\; q\, S\, C_{NN\alpha}\, \alpha \left[1 + q \;/\; (q_D - q) \right], \text{ or}$$

$$N_N \;=\; q\, S\, C_{NN\alpha}\, \alpha\, q_D \,/\, (q_D - q).$$

Thus the nose tip has a second aeroelastic increment in angle of attack. The final nose angle of attack becomes

$$\frac{\alpha_{Rigid}\; q_D \left\{ x_1\, C_{N1\alpha} - x_2\, C_{N2\alpha} - x_3\, C_{NN\alpha} \right\}}{\left\{ x_1\, C_{N1\alpha} - x_2\, C_{N2\alpha} \right\} (q_D - q) - x_3\, C_{NN\alpha}\, q_D} \quad (2)$$

The factor, $q_D \,/\, (q_D - q)$, is the second factor by which rigid body nose loads are increased due to static aeroelastic effects. To implement these in, say BENDIT, increase the angle of attack for all elements except the nose using eq. (1), and increase the nose angle of attack using eq. (2)

- The Flexible Joint Problem

Introduction

A variation on the flexible upper stage and payload problem described above is an essentially rigid vehicle except for a single flexible joint. For purposes of discussion, the joint is assumed to attach payload and rocket body.

Analysis

The analysis proceeds as before, except the payload normal force is now

$$N_N = q S C_{NN\alpha} (\alpha + \varphi),$$

where φ = the joint rotation angle

The joint rotation angle, in turn, will depend linearly on the bending moment acting on the joint:

$$\varphi = C M,$$

where C = joint compliance, and
M = bending moment acting on the joint.

The appendix provides a discussion of the factors affecting joint compliance.
Then, if the joint is a distance x_j behind the nose center of pressure

$$N_N = q S C_{NN\alpha} (\alpha + C N_N x_j), \text{ or}$$

$$N_N = q S C_{NN\alpha} \alpha / (1 - q S C_{NN\alpha} C x_j).$$

The divergence dynamic pressure q_D is

$$q_D = 1 / S C_{NN\alpha} C x_j.$$

With this the nose normal force becomes

$$N_N = S C_{NN\alpha} \alpha q_D / (q_D - q).$$

The factor to be applied to rigid body loads to obtain loads with a flexible joint is just

$q_D / (q_D - q)$, the same as for the flexible body bending case.